Cambodia: Institutional Trajectory and Outcome Decay

Tracing 1970–2025 through the cross-sectional ML model

Institutional Complexity Project

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Contents

Motivation

The cross-sectional analysis in `outcomes_ml.Rmd` answers: *given a country’s 2025 V-Dem fingerprint, what outcomes does the model expect?* The Cambodia case study answers: *what country does Cambodia institutionally resemble today?* (answer: Burundi). Both are snapshots.

This document asks the time-series question: **how has Cambodia’s institutional fingerprint evolved since 1970, and what would that historical fingerprint have predicted at each point in time about its development outcomes?** Cambodia’s GDP-per-capita series has visible breaks at 1979, 1987, 1997, and 2006; we trace whether its institutional setup tracks those breaks and compare to two benchmarks: Vietnam (a managed-state ASEAN neighbour) and Burundi (its data-driven institutional twin).

The hypothesis under test is the **decay hypothesis**: prior Cambodian institutional regimes were better at producing human capital and economic complexity than the current one. The country still has a high *stock* of human capital and infrastructure (built under the post-UNTAC era and the PRK education recovery), but the current institutional setup is less able to replenish that stock. **Infant mortality may be the leading indicator** — since IMR responds to live institutional capacity (clinic staffing, vaccine cold chain) faster than HCI does, a model-predicted IMR that has turned upward before actual IMR would suggest the next wave is already cresting.

Data assembly

2025 cross-section: feature set, normalisation bounds, models

We rebuild the same `Mcp` matrix used in `outcomes_ml.Rmd`: V-Dem 2025 component-type indicators with at least 150-country coverage, dropping `v2regopploc` and `v2regsuploc`. Each feature is min-max normalised on the 2025 cross-section to `[0, 1]`. The **per-feature min and max from 2025 are saved** — they’re the bounds we apply to the historical panel later, so a 1975 Cambodia value gets placed on the same scale the model was trained on.

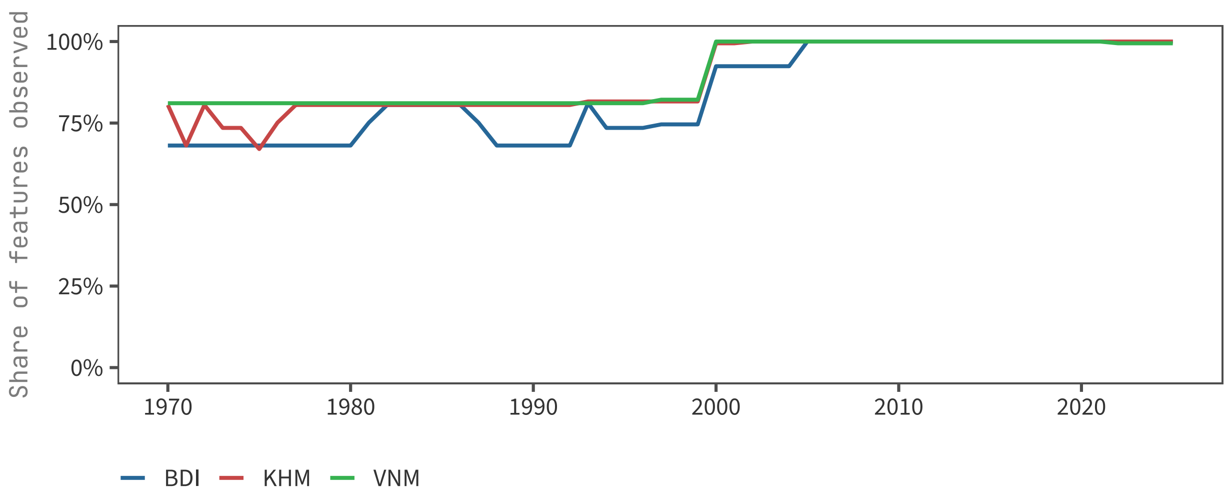
```
## V-Dem year: 2025
```

```
## Mcp: 179 countries x 185 features
```

Historical panel: KHM, VNM, BDI from 1970

For each focus country and each year in 1970–latest_year, we pull the same V-Dem feature columns, then min-max normalise using the **2025 bounds** (not the historical min/max — we want the model to see history in 2025-scale units). NAs are replaced with 0, matching the cross-sectional pipeline. NAs are common in Cambodia 1975–1999 because several V-Dem features (election quality, party characteristics, lower-chamber behaviour) are undefined when elections or legislatures don't exist; treating them as floor values is consistent with the model's training-time handling.

```
## Historical panel: 168 rows x 185 features
```




```
## wdi_hd_hci_ovrl            | best depth 3 | OOS R^2 0.771 | n 156
## wdi_sp_dyn_imrt_in         | best depth 5 | OOS R^2 0.710 | n 170
## wdi_sp_dyn_le00_in         | best depth 3 | OOS R^2 0.629 | n 171
## atl_sitc_eci               | best depth 3 | OOS R^2 0.450 | n 172
```

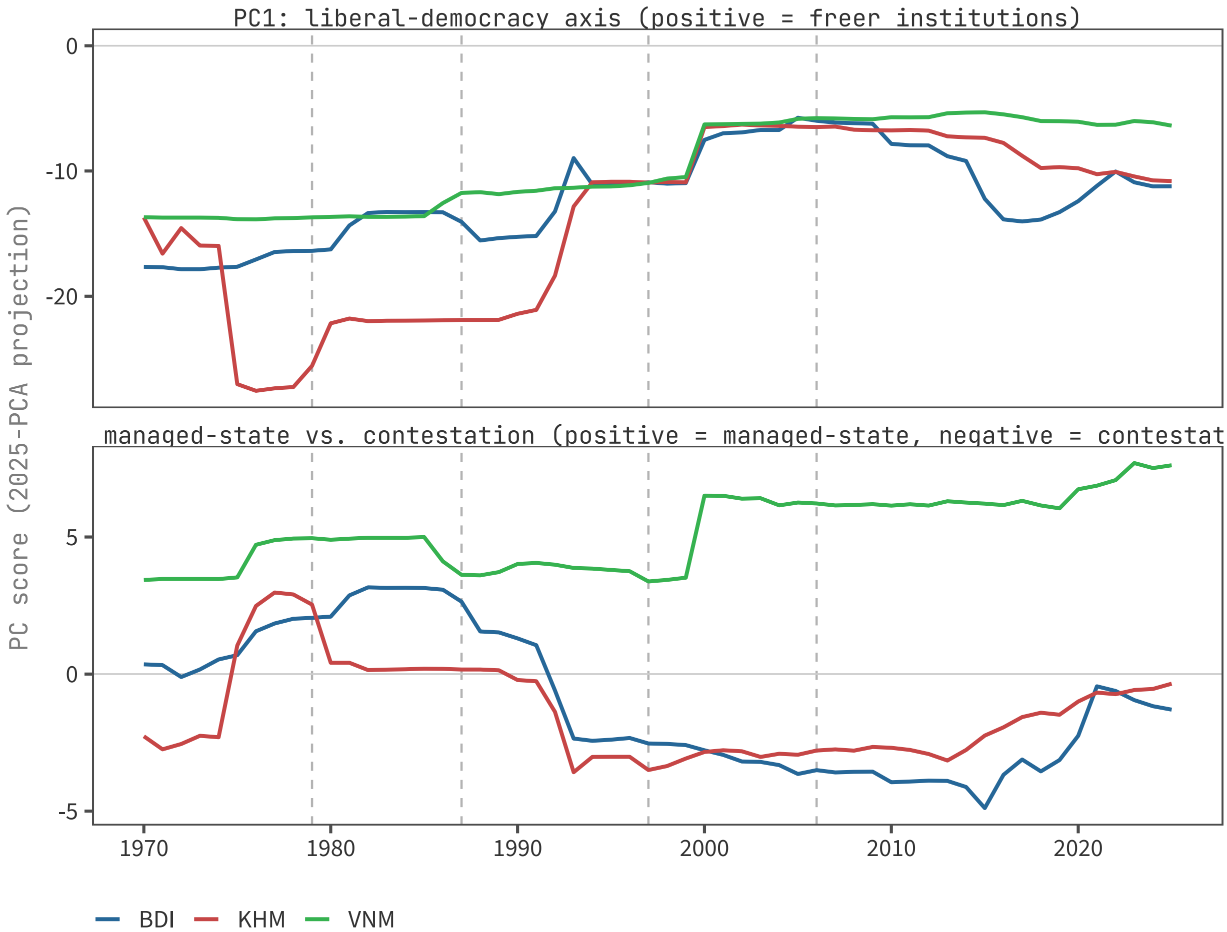
Table 1: Cross-sectional model performance per outcome (5-fold CV). These are the same models used in outcomes_ml.Rmd, here re-fit on the four outcomes we project history through.

bucket	label	n	best_depth	oos_r2	oos_rmse
Education	Human Capital Index	156	3	0.771	0.069
Health	Infant mortality (log)	170	5	0.710	0.576
Health	Life expectancy	171	3	0.629	4.434
Complexity	Economic Complexity (ECI)	172	3	0.450	0.753

Institutional trajectory, 1970–2025

Projection through the 2025 PCA

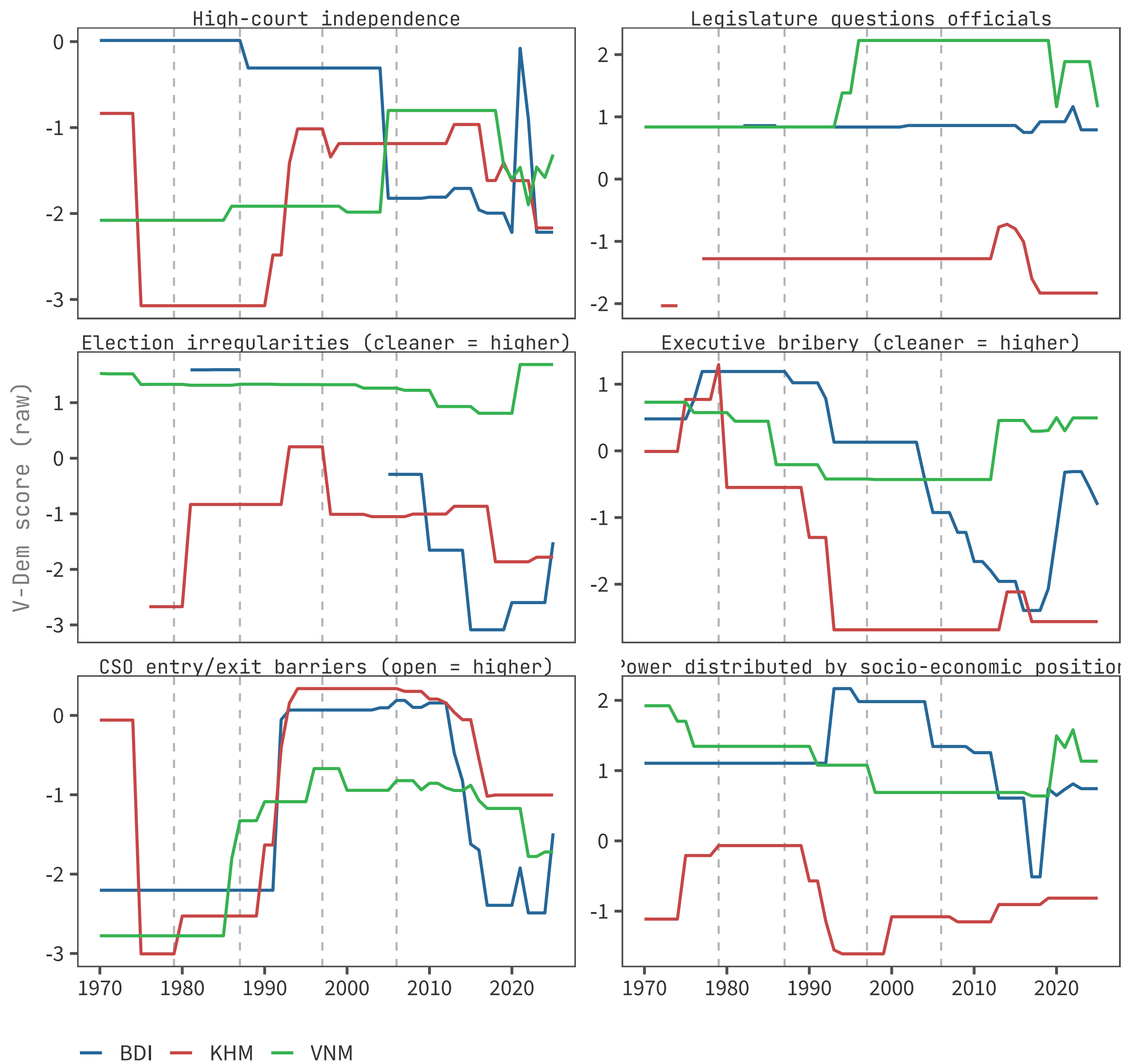
We compute `prcomp(X_mat, center=TRUE, scale•=TRUE)` on the 2025 Mcp and use its rotation matrix to project the historical panel into the same PC space. A historical PC1/PC2 score is the dot product of the historical fingerprint (2025-scaled) with the 2025 loading vector. This is the standard approach to "projecting new observations into an existing PCA."



The 2025-PCA loadings define PC1 as a liberal-democracy axis (more positive = freer judiciary, election integrity, parliamentary oversight) and PC2 as a managed-state vs. contestation axis. Top |PC2| loadings are anti-system movements (`v2csantimv`), academic/general mobilisation (`v2cademmob`, `v2cagenmob`), civil violence (`v2caviol`), and opposition camps (`v2cacamps`) on the *negative* side, versus social-media regulatory capacity (`v2smregcap`), state-driven media frames (`v2smmefra`, `v2smpolsoc`), public-health and education-equality delivery (`v2pehealth`, `v2peedueq`), and election-ban scope (`v2elsnlsff`) on the *positive* side. So **negative PC2 = open contestation, mass mobilisation; positive PC2 = state-managed information environment with delivered services**. This is the same convention used in the static `outcomes_ml.`Rmd PCA scatter (where Vietnam, China, Laos, Egypt sit deep on the positive side; Cambodia, Pakistan, Burundi, Myanmar sit on the negative side).

Selected V-Dem features over time

To complement the PC summary, we plot a small set of substantive indicators that load heavily on PC1. Each panel shows the *raw* V-Dem score (not 2025-normalised) so the y-axis stays interpretable in V-Dem's native units.



Cambodia's eras at a glance

Years	Regime	Headline institutional fact
1970–1975	Khmer Republic (Lon Nol)	Coup against Sihanouk; civil war
1975–1979	Democratic Kampuchea (Khmer Rouge)	Total state collapse; mass atrocity
1979–1989	People's Republic of Kampuchea	Vietnamese-backed PRK; rebuild from zero
1989–1993	State of Cambodia / UNTAC	Transition; UN-supervised election
1993–1997	Constitutional monarchy / coalition	FUNCINPEC–CPP power-sharing
1997–2017	CPP-dominated multiparty	Hun Sen consolidates after 1997 coup

Years	Regime	Headline institutional fact
2017–2025	One-party CPP	CNRP dissolved; Hun Manet succession

The growth breaks the user identified line up with these transitions: **1979** ends the Khmer Rouge; **1987** sits inside the PRK reform period (party reform, gradual market opening, Vietnamese drawdown announced); **1997** is the Hun Sen coup that ended power-sharing; **2006** roughly marks the consolidation of CPP dominance and accession to the WTO (2004) plus the subsequent garment-export boom.

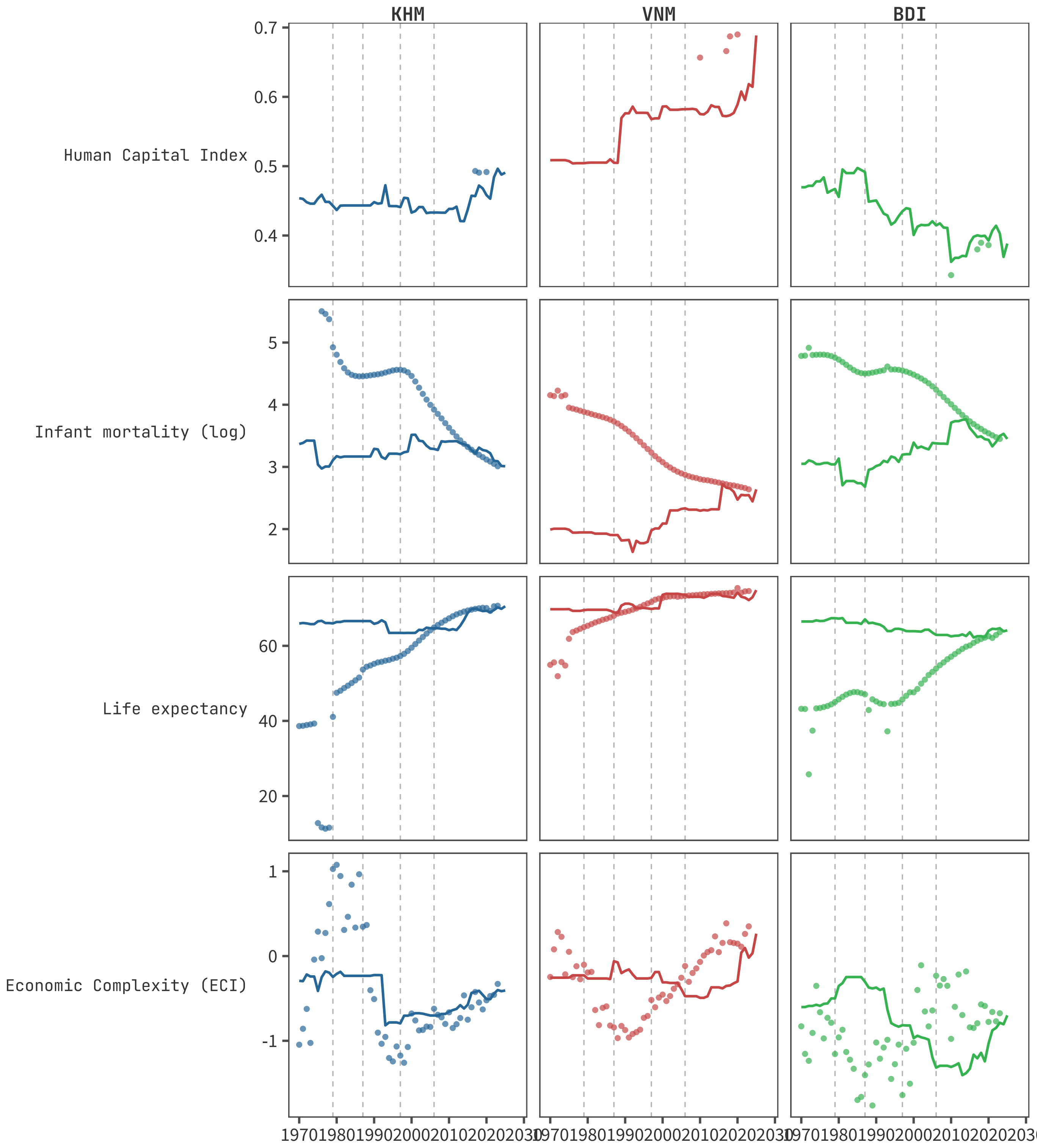
Predicted outcomes from historical fingerprints

For each year and country, we feed the 2025-scale historical fingerprint through each of the four cross-sectional XGBoost models. The result is a **counterfactual outcome time-series**: "if a country with this V-Dem fingerprint existed in 2025, this is the HCI / IMR / LE / ECI it would have." We then plot this against the *actual* outcome series (where available) and look for divergence.

The prediction is not "what HCI Cambodia had in 1985"; HCI as a measure didn't exist then. It's **"what HCI a country with 1985-Cambodia's institutional setup would have today"** — i.e., what the model thinks 1985-Cambodia's institutions could *deliver* if measured by today's yardsticks.

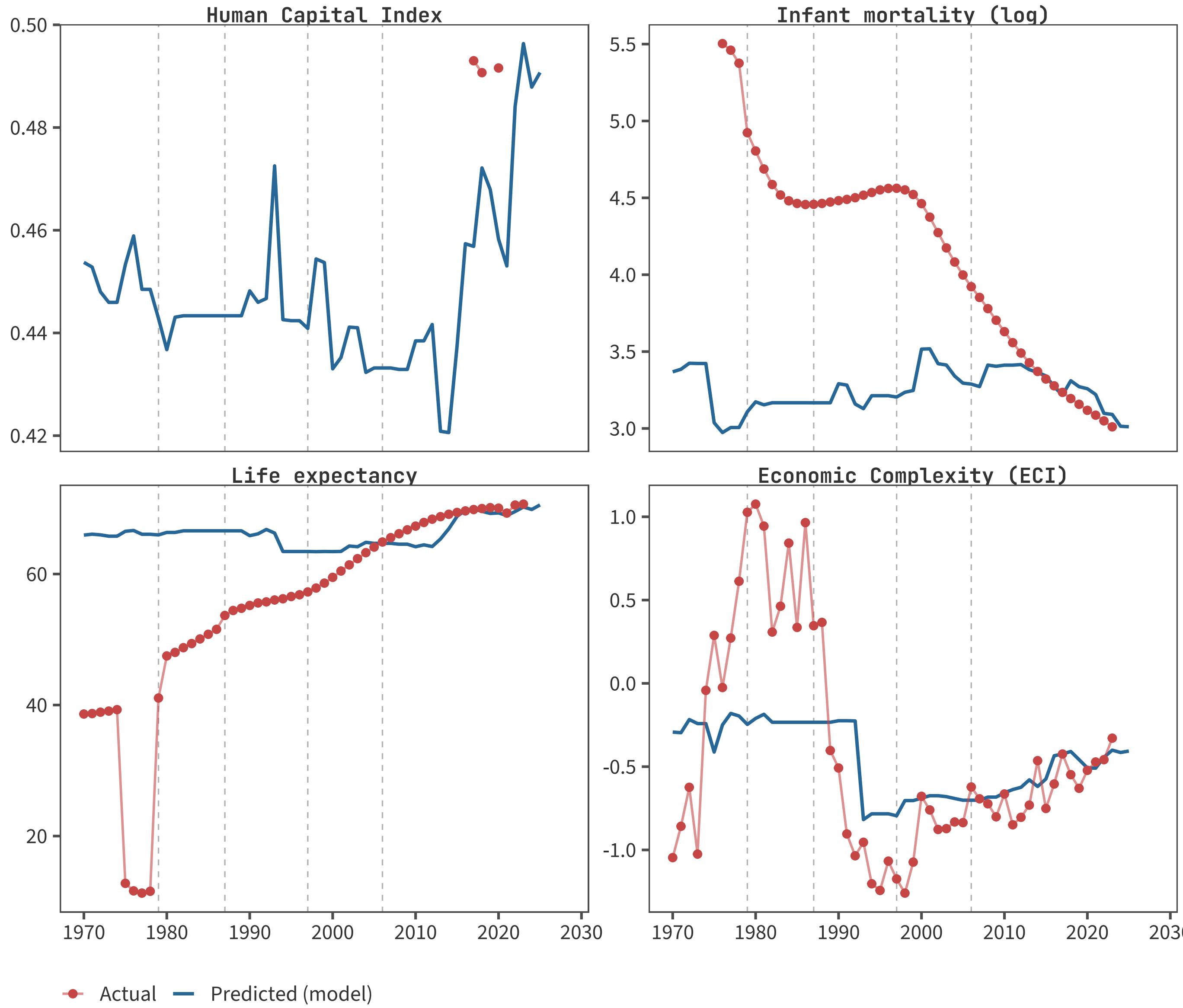
Predicted vs actual: full panel

Solid line = model prediction from the year-t V-Dem fingerprint. Dots = actual observed outcome (when available). Vertical dashes mark Cambodia's growth breaks. For HCI, actuals only exist 2010–2020 — so the dot cluster is small but informative.



Cambodia: predicted-vs-actual zoom, by outcome

A focused view of Cambodia alone makes the divergences easier to read.



Reading each panel

Human Capital Index

HCI is reported only every 2–3 years, starting 2010. The model predicts an HCI trajectory all the way back to 1970 because the V-Dem fingerprint exists for those years. This is the trajectory under test for the decay hypothesis: did the model think Cambodia *could* deliver higher HCI under earlier institutions than it predicts today?

Table 2: Predicted HCI from year-t V-Dem fingerprint vs. actual HCI (where reported).

iso3	year	predicted	actual
KHM	1975	0.453	NA
KHM	1980	0.437	NA

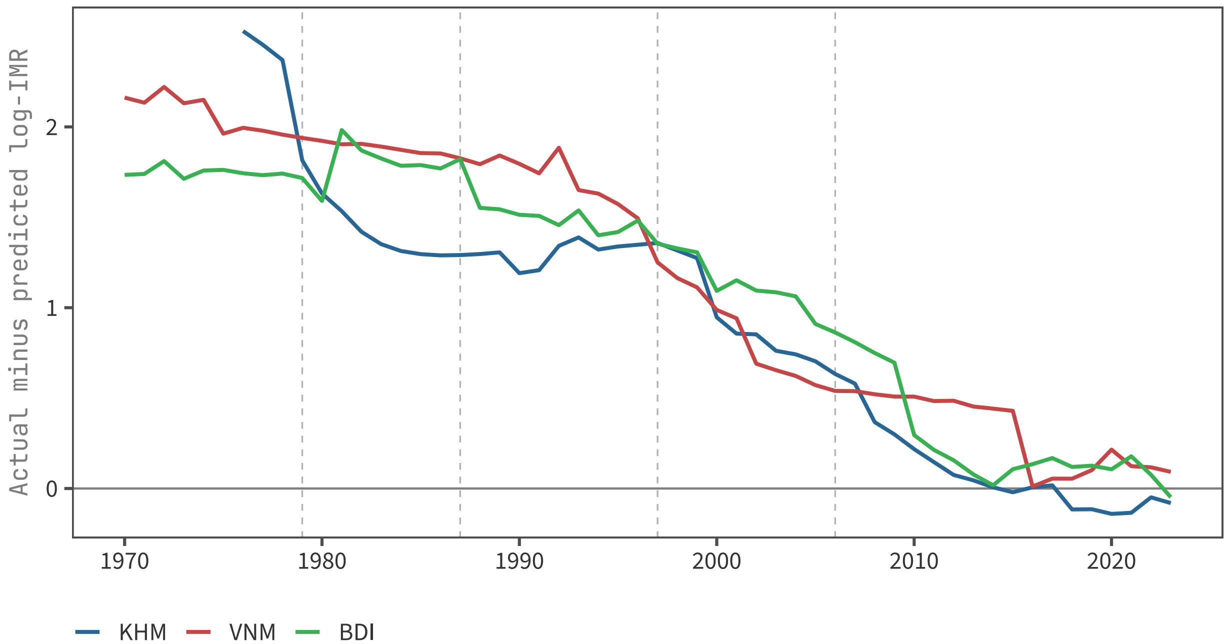
iso3	year	predicted	actual
KHM	1985	0.443	NA
KHM	1990	0.448	NA
KHM	1995	0.442	NA
KHM	2000	0.433	NA
KHM	2005	0.433	NA
KHM	2010	0.438	NA
KHM	2015	0.438	NA
KHM	2020	0.458	0.492
KHM	2024	0.488	NA
VNM	1975	0.507	NA
VNM	1980	0.505	NA
VNM	1985	0.505	NA
VNM	1990	0.576	NA
VNM	1995	0.577	NA
VNM	2000	0.586	NA
VNM	2005	0.582	NA
VNM	2010	0.575	0.657
VNM	2015	0.585	NA
VNM	2020	0.589	0.690
VNM	2024	0.614	NA
BDI	1975	0.478	NA
BDI	1980	0.456	NA
BDI	1985	0.497	NA
BDI	1990	0.451	NA
BDI	1995	0.419	NA
BDI	2000	0.401	NA
BDI	2005	0.421	NA
BDI	2010	0.362	0.343
BDI	2015	0.389	NA
BDI	2020	0.393	0.386
BDI	2024	0.369	NA

Infant mortality (log) — the leading-indicator hypothesis

Actual log-IMR for Cambodia falls steadily from 1980 onward — the HCI investment of the PRK and post-UNTAC eras delivered a real mortality transition. The model's *predicted* log-IMR is what 2025-trained institutions-to-IMR mapping says a country with year-t Cambodia's institutions would have today. If that predicted line **turns upward before actual**, it's evidence the institutions capable of *sustaining* the mortality transition have already eroded.

```
## Cambodia: predicted log-IMR minimum at year 1976.
```

```
## Cambodia: actual log-IMR minimum at year 2023 (latest available).
```



Life expectancy — the stock indicator

Life expectancy is an integral of past mortality conditions; it should respond *more slowly* than IMR and so should track actual better than predicted in the near term, even if predicted is already deteriorating.

Economic complexity (ECI)

ECI is computed from export structure, which itself is built up over decades from accumulated productive capabilities. Cambodia's ECI rose with garment exports post-WTO accession (2004); the question is whether the institutional fingerprint is now consistent with continued upgrading or has decayed below the level associated with sustained complexity gains.

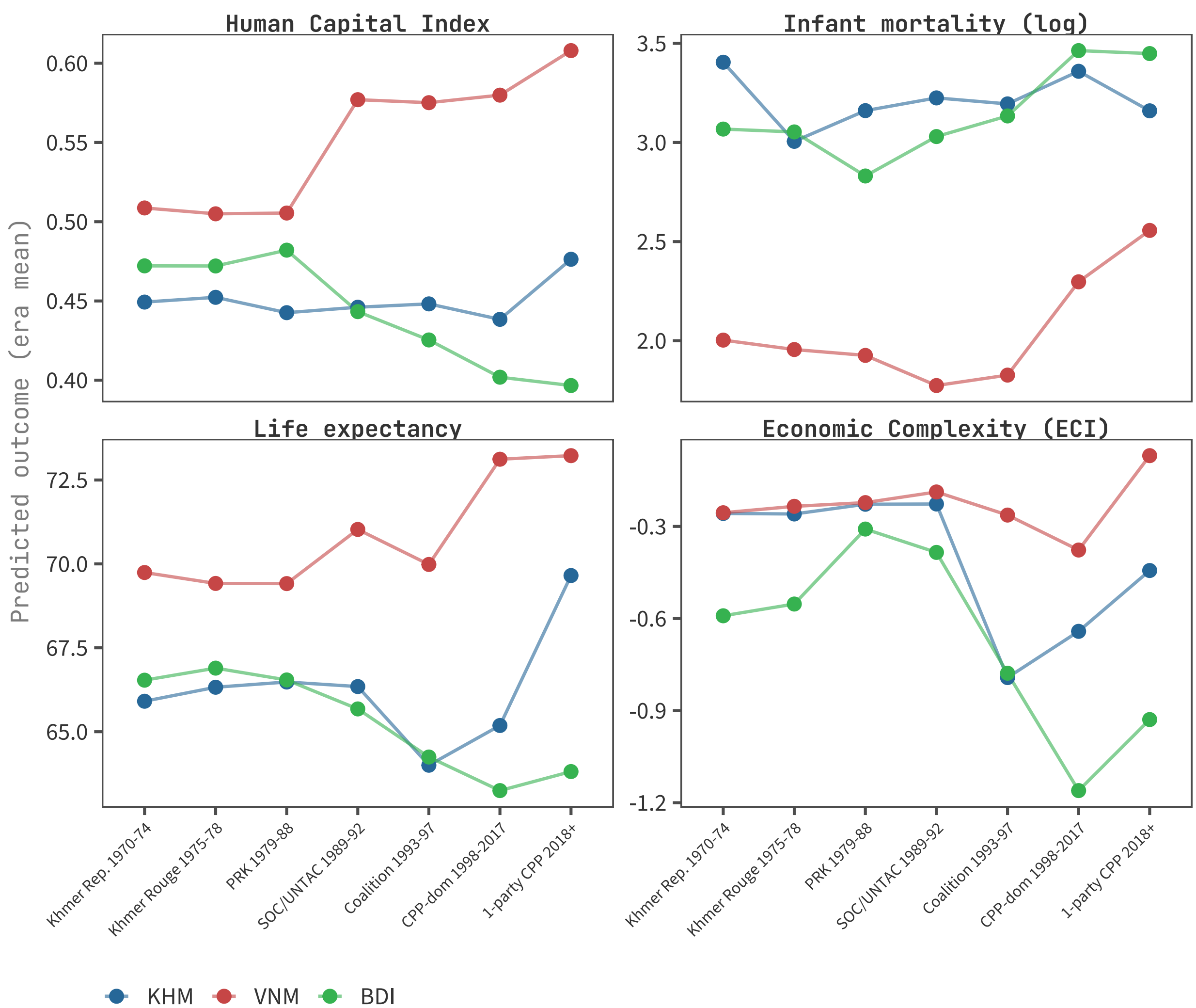
Per-era summary

To compare Cambodia's institutional regimes directly, we average each country's predicted and actual outcome within seven Cambodian eras: Khmer Republic, Khmer Rouge, PRK, SOC/UNTAC, Coalition, CPP-dominated, and One-party CPP. The Vietnam and Burundi columns use the same year ranges so the comparison is era-anchored on Cambodia's politics.

Table 3: Cambodia: era-mean predicted vs actual outcome.

Era	HCI (Predicted)	log-IMR (Predicted)	LE (Predicted)	ECI (Predicted)	HCI (Actual)	log-IMR (Actual)	LE (Actual)	ECI (Actual)
Khmer Rep. 1970-74	0.45	3.40	65.91	-0.26	NA	NA	38.92	-0.72

Era	HCI (Predicted)	log-IMR (Predicted)	LE (Predicted)	ECI (Predicted)	HCI (Actual)	log-IMR (Actual)	LE (Actual)	ECI (Actual)
Khmer Rouge 1975-78	0.45	3.01	66.33	-0.26	NA	5.45	11.82	0.29
PRK 1979-88	0.44	3.16	66.48	-0.23	NA	4.58	49.53	0.67
SOC/UNTAC 1989-92	0.45	3.22	66.35	-0.23	NA	4.49	55.33	-0.71
Coalition 1993-97	0.45	3.19	64.00	-0.79	NA	4.55	56.58	-1.13
CPP-dom 1998-2017	0.44	3.36	65.19	-0.64	0.49	3.85	65.06	-0.77
1-party CPP 2018+	0.48	3.16	69.65	-0.44	0.49	3.10	70.11	-0.49



Structural transformation, 1970-2025

The institutional trajectory above tells us about *politics*. To diagnose whether Cambodia's industrial coalition is rentier or magician (Pritchett, Sen, Werker, *Deals and Development*, 2018), we also need the *economic* trajectory: when did GDP-per-capita break, what sectors drove each phase, and does industrial growth lift the rest of the economy or run as an enclave?

This section ports the structural-breaks and sectoral-decomposition logic from `~/dev/political-economy-pakistan/struc_breaks.Rmd` (which already renders a Cambodia PDF) onto the GL theme so they sit alongside the institutional analysis.

Growth breaks (Bai-Perron + Kar et al., 2013)

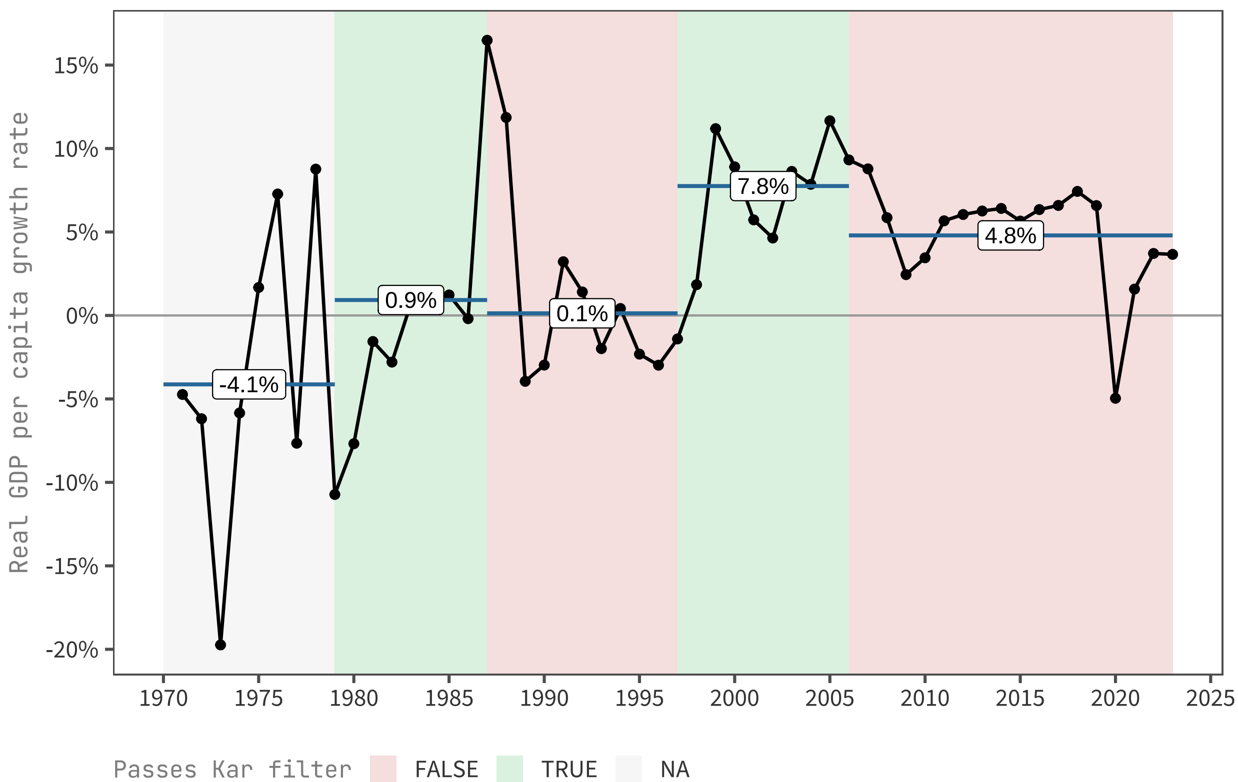


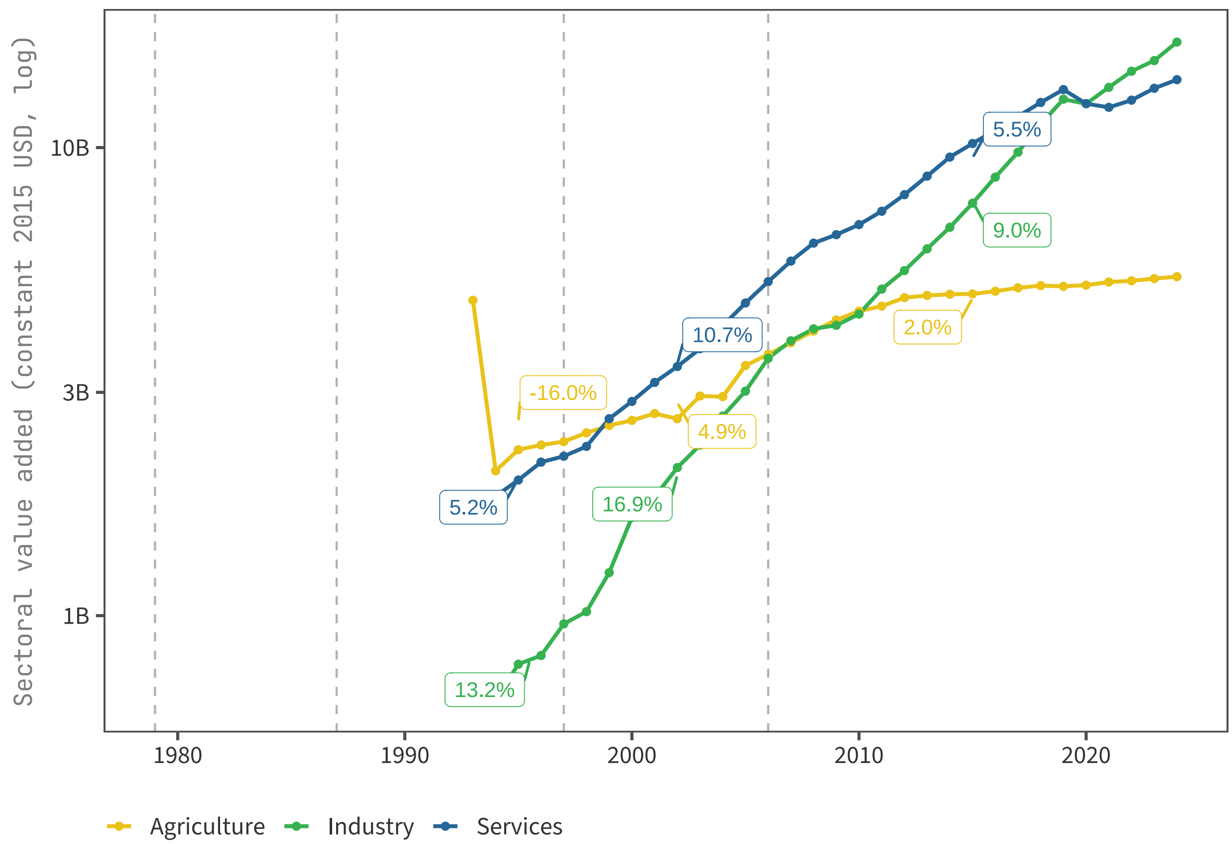
Table 4: Cambodia GDP-per-capita growth periods (Penn World Tables 11.0).

Period	Years	Avg. growth	Median growth
(1970,1979]	9	-4.1%	-5.8%
(1979,1987]	8	0.9%	0.3%
(1987,1997]	10	0.1%	-1.7%
(1997,2006]	9	7.8%	8.6%
(2006,2023]	17	4.8%	5.9%
NA	1	NA	NA

The Bai-Perron procedure picks the same four break years the analysis was seeded with: **1979, 1987, 1997, 2006**.

The Kar filter (3% absolute jump threshold for direction-changes, 1% for direction-continuations) marks 1979-87 (PRK recovery) and 1997-2006 (post-coalition garment boom) as the two genuine accelerations. The 1987-97 and 2006-23 periods are flagged as non-accelerations: the first because PRK gains stalled under sanctions and the 1991-93 transition shock; the second because growth slowed from 7.8% to 4.8% as the post-2000 garment-led boom exhausted its initial extensive margin.

Sectoral value-added trajectory



The sectoral picture sharpens the political trajectory. **Agriculture slows sharply after 2006** to ~2.0% total-VA growth — barely above Cambodia's ~1.3% population growth, so per-capita agricultural value-added is approximately flat (it was ~3.6% per-capita over 1997-2006 in the struc_breaks per-capita series). **Services slow** from 10.7% (1997-2006) to 5.5% (2006-23). **Industry holds 9.0%** as the dominant engine, but its growth rate *also* fell from the 16.9% peak of 1997-2006. The structural-transformation moment Cambodia boasts — industry overtaking services in value-added around 2017 — happened with industry slowing, not services stagnating: a level-crossing driven by services losing momentum rather than by industry accelerating. (The CAGRs labelled on the chart are total VA growth, not per-capita; subtract ~1.3pp for per-capita rates.)

Magician or rentier? Sectoral evidence

The Pritchett-Sen-Werker framework distinguishes growth that **deepens productive capability** (magician) from growth that **scales existing rent flows** (rentier). For Cambodia's industrial coalition — garments, SEZs from 2005, casinos, real estate — three diagnostic tests below operate on data we already have. A fourth test connects PC2's contestation trajectory to the political economy of consolidation.

Test 1: Capability deepening (ECI trajectory)

Magicians complexify their export basket: ECI rises as the country moves into denser, more sophisticated products. Rentiers don't: ECI stays flat or falls as growth piles into the same narrow basket.

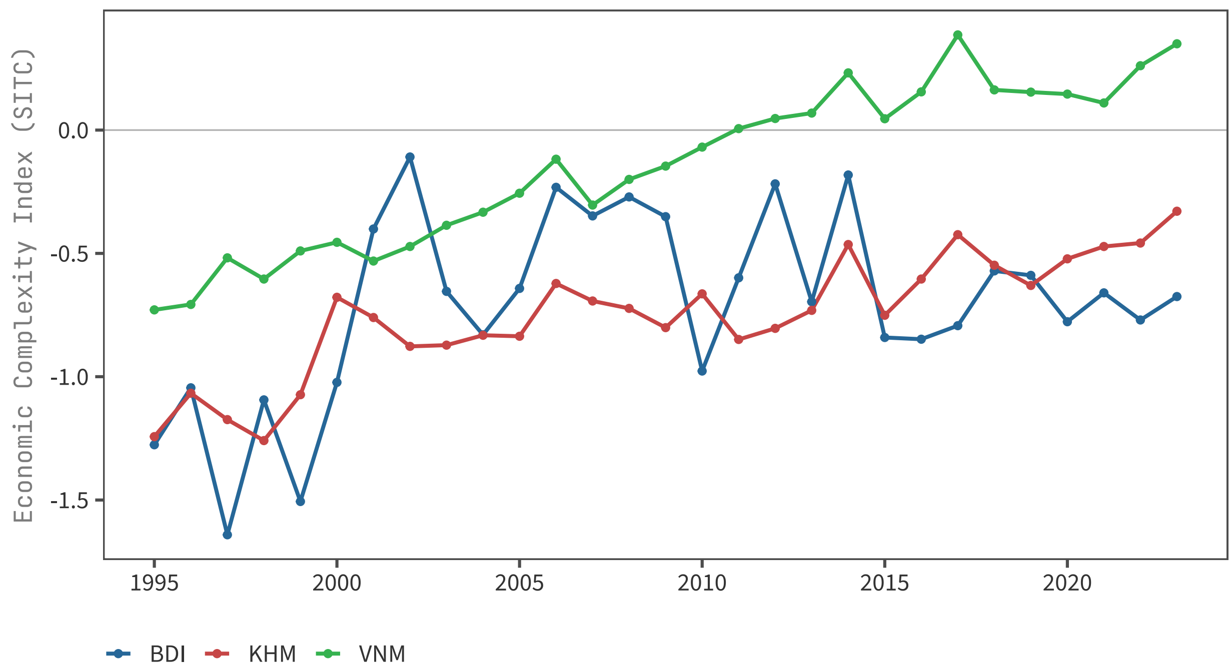


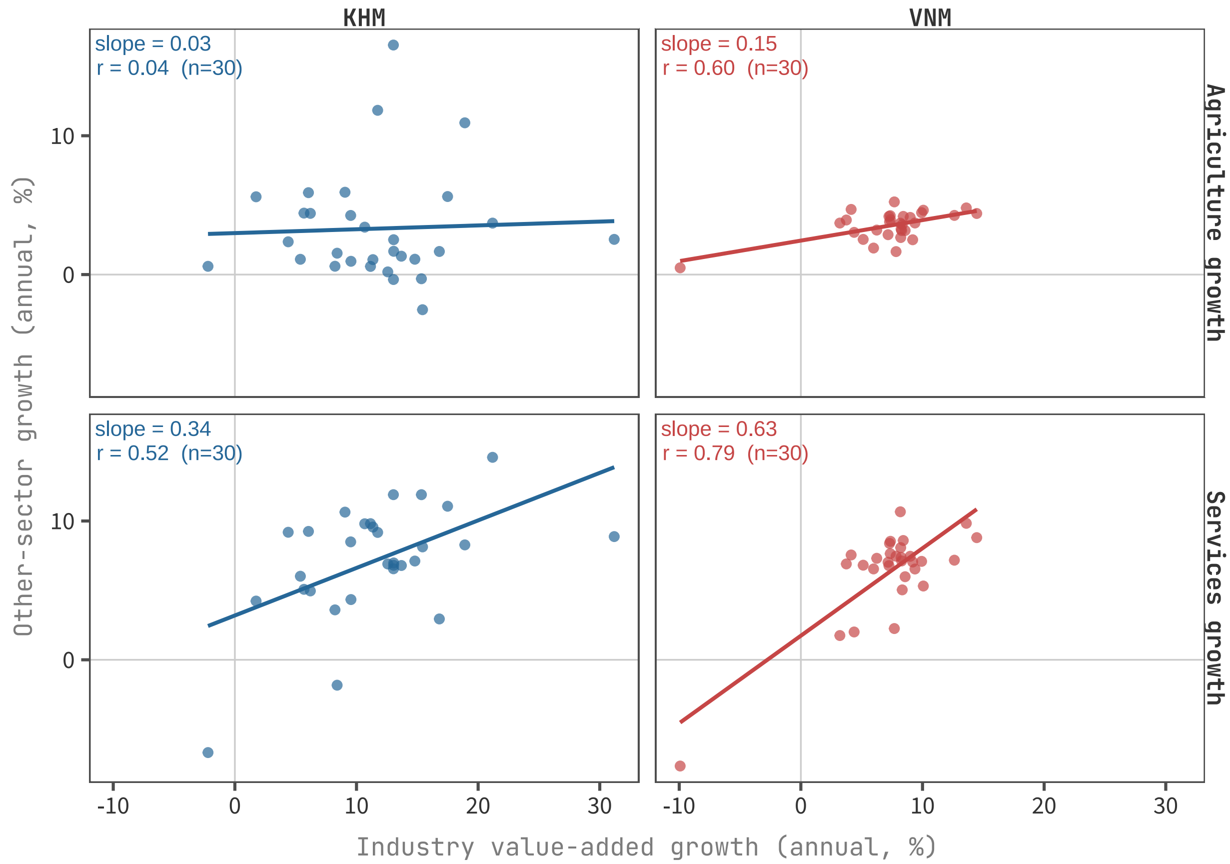
Table 5: ECI start/end and total change by country.

iso3	First year	Last year	ECI start	ECI end	ECI change
VNM	1995	2023	-0.73	0.35	1.08
KHM	1995	2023	-1.24	-0.33	0.91
BDI	1995	2023	-1.28	-0.68	0.60

The split is clean. **Vietnam** rises from a low-negative ECI in the mid-1990s to noticeably positive territory by the 2020s — a multi-decade complexification consistent with the magician profile. **Cambodia** moves modestly from very low to slightly less low; ECI has *risen* but stays firmly negative throughout, and growth in complexity has not kept pace with growth in GDP. **Burundi** drifts further negative — low complexity to begin with, slowly declining. On this test, KHM is closer to the rentier end of the spectrum than the magician end — particularly when benchmarked against VNM, which started from a similar position in 1995.

Test 2: Inter-sectoral spillovers

Magician industrial growth has linkages: a rising manufacturing sector absorbs labour from agriculture, demands transport and finance services, and pushes mechanisation back into agriculture. Rentier industrial growth is enclaved (SEZs, FDI assembly with imported inputs); other sectors don't co-move with it. We test this directly: does annual industry growth predict annual agriculture or services growth, KHM vs VNM?



Vietnam's industrial growth correlates positively with both agriculture and services growth — VNM's industry pulls the rest of the economy along. Cambodia's correlations are far weaker; KHM's industry can grow strongly in years when agriculture is flat or falling. The decoupling is exactly what you'd expect of an enclaved coalition: garment factories and SEZs import their inputs (fabric from China, machinery from Korea), assemble for export, and the value-added that stays in-country is wages plus land/concession rents — neither of which feeds backward into domestic agriculture or generates much downstream demand for sophisticated services.

Test 3: Productivity vs employment in industry

Manufacturing/industry value-added growth decomposes into a productivity component (VA per worker) and an employment component (number of workers). Magicians grow the productivity component; rentiers grow the employment component, hiring more workers at flat productivity in the quota-and-cheap-labour assembly model.

We use industry-level employment (the macro parquet has no manufacturing-only employment series): industry employment share `wdi_sl_ind_empl_zs` × total labour force `wdi_sl_tlf_totl_in`. For Cambodia, industry is dominated by garments, so the proxy is reasonable; for Vietnam, industry includes more heavy-industry and construction, so the productivity component is biased *upward* relative to garments.

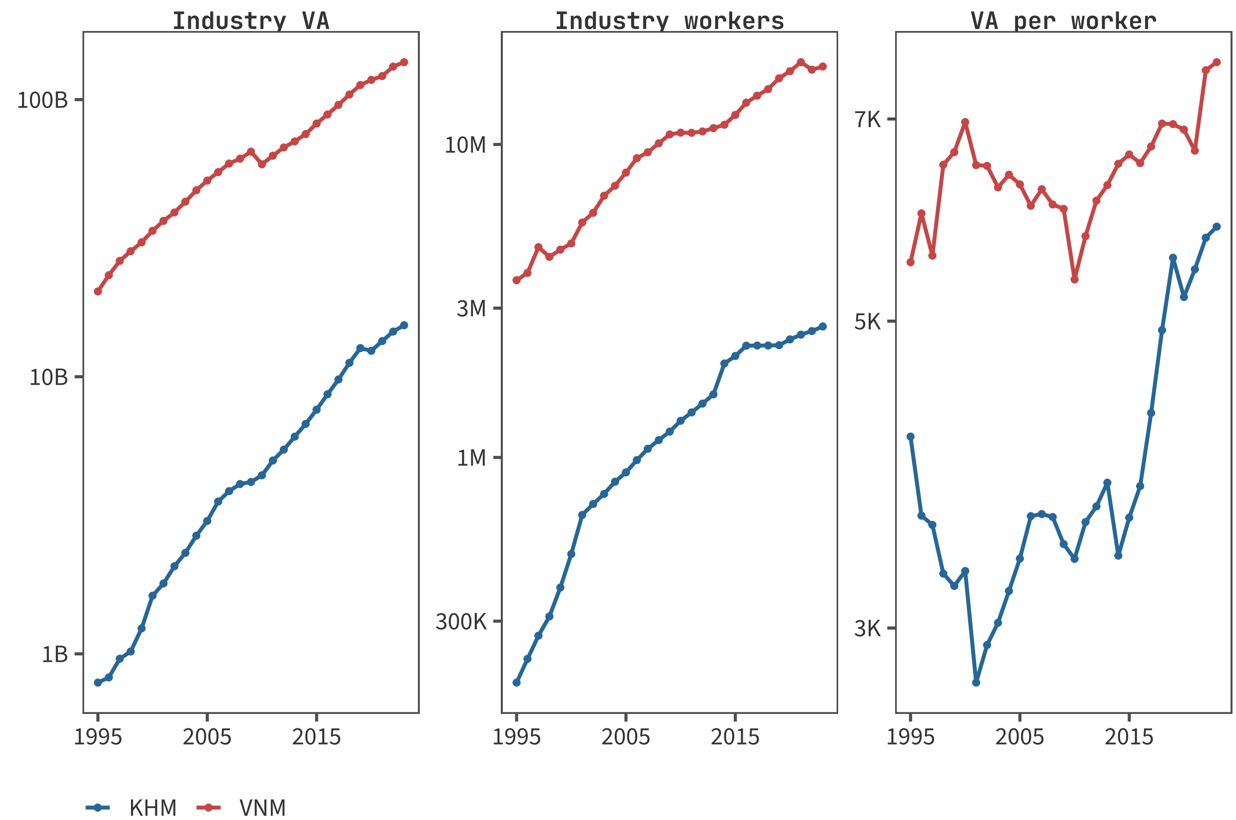


Table 6: Industry productivity vs employment decomposition, KHM vs VNM.

iso3	Period	Years	Industry VA CAGR	Workers CAGR	VA/worker CAGR
KHM	1995-2023	28	11.2%	9.8%	1.3%
VNM	1995-2023	28	7.0%	5.8%	1.2%

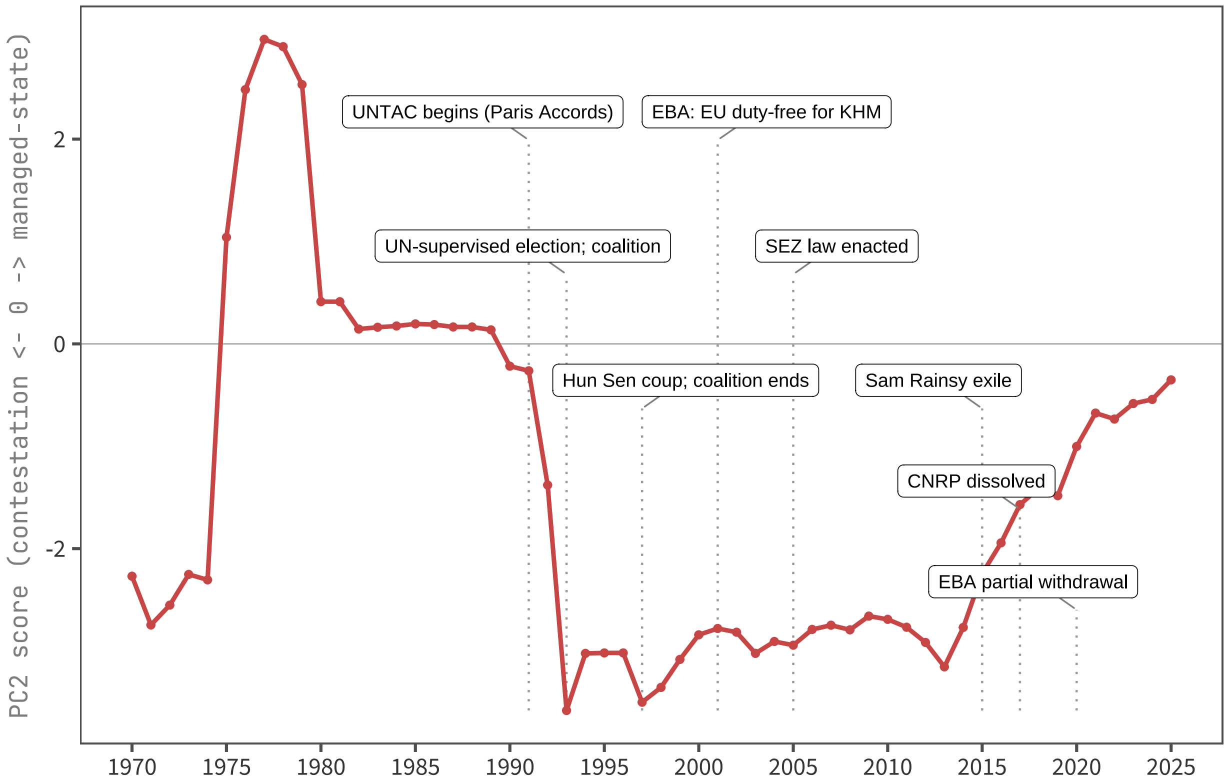
The decomposition tells a more nuanced story than the simple "rentiers grow workers, magicians grow productivity" framing predicts:

- **Productivity CAGRs are roughly equal**: KHM 1.3%/yr vs VNM 1.2%/yr. Both countries are running on the same productivity treadmill in industry — neither is a runaway productivity story.
- **Workers grew much faster in KHM** (9.8%/yr vs 5.8%/yr) — Cambodia absorbed industrial labour ~70% faster than Vietnam at a *similar* productivity-growth rate, meaning **a larger share of KHM's industry growth came from the extensive margin** (12% from productivity vs 17% for VNM).
- **Levels still differ**: VNM ended 2023 at ~$7K VA per industry worker; KHM at ~$6K. Vietnam was ahead in 1995 and remains ahead.

So the rentier signal here is partial. KHM's productivity isn't *flat* in absolute terms (it rose 1.3%/yr), but it *is* a smaller share of total industry growth than VNM's. The clean magician/rentier dichotomy breaks down at the productivity-CAGR level; what survives is that **Cambodia's industrial expansion is much more headcount-driven than Vietnam's**, even with similar per-worker productivity gains. The rentier reading wants to see Cambodia adding cheap labour into the same low-productivity activities — and that's exactly the workers-vs- productivity split this table shows.

Test 4: PC2 trajectory and coalition consolidation

If Cambodia's industrial coalition is rentier-style, the political economy story is that **rising managed-state capacity (PC2 trending positive) should align with coalition-cementing events** — moments when the CPP locks in the textile/SEZ/concession bargain and forecloses challengers — not just with formal democratic-backsliding events. We overlay PC2 with the relevant political and trade-policy moments.



After discounting the 1975-1980 spike (a model artifact: NA-imputed-to-zero fingerprints during state collapse), the genuine trajectory is a steady **multi-decade climb from deep contestation toward managed-state values**:

- **PRK era (~1985-1990): mildly managed.** PC2 hovers around 0, reflecting the Vietnamese-installed administration's top-down control before the multi-party transition.
- **UNTAC and 1993 election:** PC2 falls sharply into deep-negative contestation territory (~-3) — multi-party period, refugee camps, rebel factions, and a fragile UN-supervised coalition.
- **1995-2010: PC2 stays at -2 to -3.** The deepest contestation values in the series. KR holdouts surrender 1998-99; the 1997 Hun Sen coup forcibly ends power-sharing but factional politics, opposition parties, and active civil-society mobilisation remain part of the V-Dem fingerprint through this entire window.
- **2010-2024: PC2 rises gradually from -3 toward 0.** Sam Rainsy exiled (2015), CNRP dissolved (2017), social-media regulation and one-party legalism harden — exactly the institutional features that load on positive PC2.

This is **precisely the trajectory the rentier-consolidation hypothesis predicts**. The CPP entered the EBA-and-SEZ window (2001-2005) inside a contested political space and used the rents from that window to buy out, exile, or dissolve the contestation channels feature by feature. PC2's slow climb from -3 toward 0 is the institutional signature of that buy-out playing out — not a sign of liberalisation, but of *managed-state lock-in*. The events overlaid on the chart (1997 coup, 2015 exile, 2017 CNRP dissolution, 2020 EBA partial withdrawal) cluster on the rising leg, not the falling one.

A natural follow-up — flagged in project memory — is to decompose PC3 and PC4 (roughly the same size as PC2) to separate the *contestation/mobilisation* and *regulatory/cyber-capacity* sub-axes that PC2 currently bundles together. The story is robust at the PC2 level; PC3/PC4 should sharpen which features are doing the work.

What the four tests jointly say

Test	Magician would show	What KHM shows	Verdict
1: ECI trajectory	Rising over decades	Rises modestly, stays negative	Rentier-leaning
2: Spillovers	Industry growth lifts agri/services	Weak / no co-movement	Rentier
3: Productivity vs employment	VA/worker rises sharply	Productivity CAGR matches VNM (1.3% vs 1.2%); workers grow 70% faster than VNM	Partial: headcount-driven expansion, but productivity not flat
4: PC2 + events	n/a (political-economy mapping)	Deep contestation 1993-2010 (UNTAC, factional politics, KR remnants); steady rise toward managed-state 2010-2024 (Sam Rainsy exile, CNRP dissolution, social-media regulation)	Strong — slow climb to managed-state matches CPP coalition lock-in

Vietnam — the obvious magician comparator — passes the first two economic tests cleanly: ECI rises sharply (the strongest signal in the panel), sectoral spillovers are present and strong. On productivity, the picture is closer than expected: VNM's per-worker productivity growth is 1.2%/yr versus KHM's 1.3% — essentially the same — and the *magician* signature shows up in VNM's higher *level* (~$7K vs ~$6K in 2023) and in productivity contributing a larger *share* (17% vs 12%) of total industry growth, not in a faster productivity CAGR.

Cambodia's diagnosis: **clearly rentier-leaning on capability deepening and spillovers; only partially rentier on the productivity margin** (extensive-margin growth dominates, but productivity isn't stagnant). The political-economy story (test 4) is consistent: PC2 sits at -2 to -3 (deep contestation) through the 1993-2010 window when the rentier coalition is being assembled inside a still-contested political space, then climbs steadily toward zero (managed-state) through the 2010s as the CPP exiles, dissolves, and absorbs its remaining contestation channels.

The simple "older = better" reading isn't supported

The cross-sectional model does **not** predict that earlier Cambodian institutional setups would deliver better outcomes than the current one. For Cambodia:

- **Predicted HCI is highest under the current 2018+ one-party CPP era (0.48)** and lowest in the 1998–2017 CPP-dominated era (0.44). The Coalition era (1993–97) predicts 0.45 — between the two CPP periods, not above them.

- **Predicted log-IMR is lowest (best) in the current era (3.16)** — comparable to the PRK and Coalition eras, and lower than the CPP-dominated era's 3.36.
- **Predicted life expectancy is highest in the current era (69.7)** — almost 5 years above the long flat plateau of 64–66 that the model reads off every prior era's institutions.
- **Predicted ECI** is the partial exception: it peaks under PRK (Vietnamese-style managed state, -0.23) and SOC/UNTAC, dips deepest under the Coalition era (-0.79), and only partially recovers under current institutions (-0.44).

The Khmer Rouge era's "high" predicted values (HCI 0.45, LE 66) are a model artifact: the V-Dem fingerprint hits the floor on so many features that XGBoost extrapolates to the closest training neighbours, all of which sit far less repressively. Treat 1975–1978 predictions as out-of-calibration.

The actual story: convergence to the institutional ceiling

A different — and arguably more interesting — pattern is visible in the predicted-vs-actual plots. For all four outcomes, **actual Cambodia has caught up to predicted Cambodia**:

Outcome	1970s actual vs predicted	2020s actual vs predicted
Life expectancy	actual far below predicted (~12 vs ~66 in 1976)	actual ~ predicted (~70 vs 70)
Infant mortality (log)	actual far above predicted (~5.4 vs ~3.2)	actual ~ predicted (~3.1 vs 3.2)
HCI	(no actual)	actual ~ predicted (0.49 vs 0.48)
ECI	actual volatile, often above predicted	actual ~ predicted (-0.5 vs -0.4)

In the 1970s–1990s, Cambodia's actual outcomes were *worse* than its institutional fingerprint predicted — the gap reflecting active state collapse, war, and the genocidal Khmer Rouge baseline. As politics stabilised post-1993, the gap closed: actuals recovered to the level that the country's institutions could plausibly sustain. **By 2024 Cambodia is sitting on its institutional ceiling on every outcome we measured.** Further improvement requires institutional improvement.

What about Vietnam and Burundi?

- **Vietnam** is the cleanest "managed-state success" case. Predicted HCI rose from 0.50 (1970s) to 0.61 (2024) — a real institutional improvement. Actual HCI 2020 = 0.69 sits **above** predicted, the same kind of overperformance pattern Cambodia had decades ago. Vietnam's institutional fingerprint is delivering managed-state features (cyber capacity, bureaucratic competence, controlled but effective service delivery) that the model rewards.
- **Burundi** is the clearest decay case in the panel. Predicted HCI fell from 0.48 (mid-1980s) to 0.37 (2024). Actual HCI 2020 = 0.39 matches predicted closely. **Burundi's institutional fingerprint has worsened, and outcomes have followed it down.**

This provides context for Cambodia's institutional twin status with Burundi (the finding from `outcomes_ml.Rmd`'s peer analysis): the two countries share a similar 2025 institutional fingerprint, but Cambodia arrived at that position via a steady multi-decade *upgrade* (post-Khmer Rouge), while Burundi arrived via institutional erosion. They look the same in the cross-section but their trajectories — and the headroom between actuals and institutional ceiling — differ.

The rentier-coalition diagnosis explains the ceiling

The four sectoral tests above suggest *why* Cambodia's institutional ceiling sits where it does. Vietnam's managed-state institutions are delivering managed-state outcomes because the coalition behind them is magician-style: rising ECI, sectoral spillovers, rising productivity per worker. Cambodia's nominally similar institutional features (the country's PC1 score is not far off Vietnam's; both are managed states) deliver less because the coalition behind them is rentier-style: garments and SEZs run as enclaves, ECI barely rises, productivity per industry worker grows slowly, agriculture and services don't co-move with industry.

This reframes the institutional-twin finding (KHM-BDI). Burundi and Cambodia look the same on V-Dem features by 2025, but Burundi arrived at that fingerprint by erosion of a once-better state, while Cambodia arrived by a steady upgrade from genocide-era zero — an upgrade *driven by* a rentier coalition that has been deliberately suppressing contestation features (the negative-PC2 axis: anti-system mobilisation, opposition camps, civil violence) and substituting state regulatory capacity to protect its rents. The ceiling for both countries looks similar from above; the political economy that enforces it is opposite.

On the leading-indicator hypothesis

Strictly, the leading-indicator framing posed at the start — "predicted IMR turns up before actual" — does not appear in the data: predicted log-IMR for Cambodia is roughly flat at ~3.2 across 1970–2024, with no clear deterioration. Whatever institutional decay shows up in specific V-Dem features (election integrity, executive bribery, CSO space, all of which peaked 1995–2005 and declined since 2010) is **masked at the predicted-outcome level by simultaneous improvements in administrative-capacity features** (cyber capacity, regulation, basic services delivery) that the model also rewards.

In other words: the model reads "Cambodia 2025" as a *competent soft-authoritarian* fingerprint, not a *decayed-democratic* one. The features that have decayed (contestation, accountability) are the same features the model finds weakly predictive of IMR/HCI/LE for this class of country. So if those features have decayed without dragging the predicted outcomes down, it suggests the model thinks they aren't the binding constraint.

That doesn't mean the user's instinct is wrong — it means **the binding constraint on Cambodia's next wave of human-capital and complexity gains is institutional improvement on the features the model *does* weight heavily**: judicial independence, election integrity, executive oversight, parliamentary functionality. Those are the features whose peak-and-decline the V-Dem trajectory plots show most clearly. The predicted ceiling won't move up until those features do.

Caveats

- **The model is cross-sectional and trained on 2025.** Projecting historical fingerprints through it assumes the *mapping* from institutions to outcomes has been stable over 50 years — which is a strong assumption. Returns to particular institutional features (e.g. parliamentary oversight) may have changed over time as global development conditions changed.
- **Min-max bounds use only the 2025 cross-section.** A 1975 V-Dem score more extreme than any 2025 country falls outside the [0, 1] range that XGBoost saw in training, and the model extrapolates. This will mostly affect Khmer Rouge–era predictions; treat that period as outside the model's calibration.
- **NAs are imputed to feature minimums.** When V-Dem couldn't code a feature because the institution didn't exist (no elections, no opposition press, no legislature), our pipeline sets the feature to its 2025 minimum. This is consistent with the cross-sectional pipeline but means predictions for Cambodia 1975–1991 partially reflect that imputation choice.
- **HCI actuals only exist 2010–2020.** The "predicted HCI under 1985 institutions" cannot be validated directly;

it's a counterfactual.

- **ECI depends on export structure.** Cambodia's ECI rise post-2004 is largely driven by the garment sector, which is itself a product of trade preferences and FDI rather than the deep institutional features V-Dem captures. Any predicted-vs-actual divergence on ECI is partly a story about trade policy, not institutions per se.
- **No causal claim.** This is a descriptive exercise mapping V-Dem fingerprints onto outcome predictions. Whether earlier Cambodian institutions *caused* better outcomes — or whether the institutions themselves were a downstream symptom of prior conditions — is not identified here.
- **Industry employment is a proxy for manufacturing employment.** The WDI macro parquet has no manufacturing-only employment series, so the productivity-vs-employment decomposition uses industry-level employment (`wdi_sl_ind_empl_zs` x `wdi_sl_tlf_totl_in`). For Cambodia, industry is dominated by garments, so the proxy is close. For Vietnam, industry includes more heavy industry and construction, so VNM's VA-per-worker is biased upward relative to a pure manufacturing comparison. The qualitative split (VNM grows productivity, KHM grows headcount) is robust to this.
- **Magician-vs-rentier diagnosis is suggestive, not definitive.** A fuller test would add manufacturing subsector concentration over time (incomplete for KHM) and preference-dependence on EBA / GSP (requires HS-6 atlas trade data). Both are flagged for follow-up.